

$$\begin{aligned}
& y_d^{i1i2} \left(-\frac{1}{16\pi^2} \left(-\frac{1}{2} \overline{C_{\phi1\phi2}} y_d^{i1i2} \overline{a_e^{pr}} a_e^{pr} LF_{3,1,0} [m_l^p, m_e^r] + \frac{3}{2} \overline{C_{\phi1\phi2}} y_d^{i1i2} \overline{a_e^{pr}} a_e^{pr} LF_{4,1,-1} [m_l^p, m_e^r] - \right. \right. \\
& \overline{C_{\phi1\phi2}} y_d^{i1i2} \overline{a_e^{pr}} a_e^{pr} LF_{5,1,-2} [m_l^p, m_e^r] - \frac{3}{2} \overline{C_{\phi1\phi2}} y_d^{i1i2} \overline{a_d^{pr}} a_d^{pr} LF_{3,1,0} [m_q^p, m_d^r] + \\
& \frac{9}{2} \overline{C_{\phi1\phi2}} y_d^{i1i2} \overline{a_d^{pr}} a_d^{pr} LF_{4,1,-1} [m_q^p, m_d^r] - 3 \overline{C_{\phi1\phi2}} y_d^{i1i2} \overline{a_d^{pr}} a_d^{pr} LF_{5,1,-2} [m_q^p, m_d^r] - \\
& \frac{3}{2} \overline{C_{\phi1\phi2}} \tilde{\mu}^2 y_d^{i1i2} \overline{y_u^{pr}} y_u^{pr} LF_{3,1,0} [m_u^r, m_q^p] + \frac{9}{2} \overline{C_{\phi1\phi2}} \tilde{\mu}^2 y_d^{i1i2} \overline{y_u^{pr}} y_u^{pr} LF_{4,1,-1} [m_u^r, m_q^p] - \\
& 3 \overline{C_{\phi1\phi2}} \tilde{\mu}^2 y_d^{i1i2} \overline{y_u^{pr}} y_u^{pr} LF_{5,1,-2} [m_u^r, m_q^p] + \overline{C_{\phi1\phi2}} g_1^2 y_d^{i1i2} LF_{3,1,-1} [\tilde{\mu}, m_1] - \\
& 2 \overline{C_{\phi1\phi2}} g_1^2 y_d^{i1i2} LF_{4,1,-2} [\tilde{\mu}, m_1] + \overline{C_{\phi1\phi2}} g_1^2 y_d^{i1i2} LF_{5,1,-3} [\tilde{\mu}, m_1] + \\
& 3 \overline{C_{\phi1\phi2}} g_2^2 y_d^{i1i2} LF_{3,1,-1} [\tilde{\mu}, m_2] - 6 \overline{C_{\phi1\phi2}} g_2^2 y_d^{i1i2} LF_{4,1,-2} [\tilde{\mu}, m_2] + \\
& 3 \overline{C_{\phi1\phi2}} g_2^2 y_d^{i1i2} LF_{5,1,-3} [\tilde{\mu}, m_2] + \frac{1}{9} m_1 \tilde{\mu} g_1^2 y_d^{i1i2} LF_{1,1,1,0} [m_1, m_d^{i2}, m_q^{i1}] - \\
& \frac{1}{36} m_1 g_1^2 (\overline{C_{\phi1\phi2}} a_d^{i1i2} + C_{\phi1^2} \tilde{\mu} y_d^{i1i2}) LF_{2,2,1,-1} [m_1, m_d^{i2}, m_q^{i1}] + \\
& \frac{1}{3} m_1 \tilde{\mu} g_1^2 y_d^{i1i2} LF_{1,1,1,0} [m_1, m_d^{i2}, \tilde{\mu}] - \frac{1}{36} m_1 g_1^2 (\overline{C_{\phi1\phi2}} a_d^{i1i2} + C_{\phi1^2} \tilde{\mu} y_d^{i1i2}) \\
& LF_{2,2,1,-1} [m_1, m_q^{i1}, m_d^{i2}] + \frac{1}{6} m_1 \tilde{\mu} g_1^2 y_d^{i1i2} LF_{1,1,1,0} [m_1, m_q^{i1}, \tilde{\mu}] - \\
& \frac{1}{6} \overline{C_{\phi1\phi2}} g_1^2 y_d^{i1i2} LF_{2,2,1,-2} [m_1, \tilde{\mu}, m_d^{i2}] + \frac{1}{6} m_1 C_{\phi1^2} \tilde{\mu} g_1^2 y_d^{i1i2} LF_{2,2,1,-1} [m_1, \tilde{\mu}, m_d^{i2}] - \\
& \frac{1}{12} \overline{C_{\phi1\phi2}} g_1^2 y_d^{i1i2} LF_{2,2,1,-2} [m_1, \tilde{\mu}, m_q^{i1}] + \frac{1}{12} m_1 C_{\phi1^2} \tilde{\mu} g_1^2 y_d^{i1i2} LF_{2,2,1,-1} [m_1, \tilde{\mu}, m_q^{i1}] + \\
& \frac{3}{2} m_2 \tilde{\mu} g_2^2 y_d^{i1i2} LF_{1,1,1,0} [m_2, m_q^{i1}, \tilde{\mu}] - \frac{3}{4} \overline{C_{\phi1\phi2}} g_2^2 y_d^{i1i2} LF_{2,2,1,-2} [m_2, \tilde{\mu}, m_q^{i1}] + \\
& \frac{3}{4} m_2 C_{\phi1^2} \tilde{\mu} g_2^2 y_d^{i1i2} LF_{2,2,1,-1} [m_2, \tilde{\mu}, m_q^{i1}] - \frac{8}{3} m_3 \tilde{\mu} g_3^2 y_d^{i1i2} LF_{1,1,1,0} [m_3, m_d^{i2}, m_q^{i1}] + \\
& \frac{2}{3} m_3 g_3^2 (\overline{C_{\phi1\phi2}} a_d^{i1i2} + C_{\phi1^2} \tilde{\mu} y_d^{i1i2}) LF_{2,2,1,-1} [m_3, m_d^{i2}, m_q^{i1}] + \\
& \frac{2}{3} m_3 g_3^2 (\overline{C_{\phi1\phi2}} a_d^{i1i2} + C_{\phi1^2} \tilde{\mu} y_d^{i1i2}) LF_{2,2,1,-1} [m_3, m_q^{i1}, m_d^{i2}] + \\
& \frac{1}{18} m_1 g_1^2 (\overline{C_{\phi1\phi2}} a_d^{i1i2} + C_{\phi1^2} \tilde{\mu} y_d^{i1i2}) LF_{2,1,1,0} [m_d^{i2}, m_1, m_q^{i1}] - \\
& \frac{1}{18} m_1 g_1^2 (\overline{C_{\phi1\phi2}} a_d^{i1i2} + C_{\phi1^2} \tilde{\mu} y_d^{i1i2}) LF_{3,1,1,-1} [m_d^{i2}, m_1, m_q^{i1}] - \\
& \frac{4}{3} m_3 g_3^2 (\overline{C_{\phi1\phi2}} a_d^{i1i2} + C_{\phi1^2} \tilde{\mu} y_d^{i1i2}) LF_{2,1,1,0} [m_d^{i2}, m_3, m_q^{i1}] + \\
& \frac{4}{3} m_3 g_3^2 (\overline{C_{\phi1\phi2}} a_d^{i1i2} + C_{\phi1^2} \tilde{\mu} y_d^{i1i2}) LF_{3,1,1,-1} [m_d^{i2}, m_3, m_q^{i1}] - \\
& \tilde{\mu} y_d^{pi2} y_u^{i1r} \overline{a_u^{pr}} LF_{1,1,1,0} [m_q^p, m_u^r, \tilde{\mu}] - \frac{1}{2} \tilde{\mu} y_d^{pi2} y_u^{i1r} (C_{\phi1^2} \overline{a_u^{pr}} + \overline{C_{\phi1\phi2}} \tilde{\mu} \overline{y_u^{pr}}) \\
& LF_{2,1,1,0} [m_q^p, m_u^r, \tilde{\mu}] + \frac{1}{2} \tilde{\mu} y_d^{pi2} y_u^{i1r} (C_{\phi1^2} \overline{a_u^{pr}} + \overline{C_{\phi1\phi2}} \tilde{\mu} \overline{y_u^{pr}}) LF_{3,1,1,-1} [m_q^p, m_u^r, \tilde{\mu}] + \\
& \frac{1}{4} \tilde{\mu} y_d^{pi2} y_u^{i1r} (C_{\phi1^2} \overline{a_u^{pr}} + \overline{C_{\phi1\phi2}} \tilde{\mu} \overline{y_u^{pr}}) LF_{2,2,1,-1} [m_q^p, \tilde{\mu}, m_u^r] + \\
& \frac{1}{18} m_1 g_1^2 (\overline{C_{\phi1\phi2}} a_d^{i1i2} + C_{\phi1^2} \tilde{\mu} y_d^{i1i2}) LF_{2,1,1,0} [m_q^{i1}, m_1, m_d^{i2}] - \\
& \frac{1}{18} m_1 g_1^2 (\overline{C_{\phi1\phi2}} a_d^{i1i2} + C_{\phi1^2} \tilde{\mu} y_d^{i1i2}) LF_{3,1,1,-1} [m_q^{i1}, m_1, m_d^{i2}] - \\
& \frac{4}{3} m_3 g_3^2 (\overline{C_{\phi1\phi2}} a_d^{i1i2} + C_{\phi1^2} \tilde{\mu} y_d^{i1i2}) LF_{2,1,1,0} [m_q^{i1}, m_3, m_d^{i2}] + \\
& \frac{4}{3} m_3 g_3^2 (\overline{C_{\phi1\phi2}} a_d^{i1i2} + C_{\phi1^2} \tilde{\mu} y_d^{i1i2}) LF_{3,1,1,-1} [m_q^{i1}, m_3, m_d^{i2}] - \\
& \frac{1}{2} \tilde{\mu} y_d^{pi2} y_u^{i1r} (C_{\phi1^2} \overline{a_u^{pr}} + \overline{C_{\phi1\phi2}} \tilde{\mu} \overline{y_u^{pr}}) LF_{2,1,1,0} [m_u^r, m_q^p, \tilde{\mu}] + \\
& \frac{1}{2} \tilde{\mu} y_d^{pi2} y_u^{i1r} (C_{\phi1^2} \overline{a_u^{pr}} + \overline{C_{\phi1\phi2}} \tilde{\mu} \overline{y_u^{pr}}) LF_{3,1,1,-1} [m_u^r, m_q^p, \tilde{\mu}] + \\
& \frac{1}{4} \tilde{\mu} y_d^{pi2} y_u^{i1r} (C_{\phi1^2} \overline{a_u^{pr}} + \overline{C_{\phi1\phi2}} \tilde{\mu} \overline{y_u^{pr}}) LF_{2,2,1,-1} [m_u^r, \tilde{\mu}, m_q^p] \Big)
\end{aligned}$$